

Math140C - HW #5

Jay Ser

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1.

(a)

Briefly explain why every singleton set in $\mathbb{R}$ is Lebesgue measurable and has Lebesgue measure 0. Then, extend this result to every countable subset of $\mathbb{R}$.

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Take any $x \in \mathbb{R}$. Letting

$$A_n = (x - \frac{1}{n}, x + \frac{1}{n}), \quad (i \in \mathbb{Z}_+)$$

Recall $X \triangle Y$ for sets denotes the symmetric difference of X and Y . For each n ,

$$A_n \triangle \{x\} = (x - \frac{1}{n}, x) \sqcup (x, x + \frac{1}{n}) \tag{1}$$

The right hand side of (1) is a finite union of intervals, so $A_n \triangle \{x\} \in \mathcal{E}$. Hence $m^*(A_n \triangle \{x\}) = m(A_n \triangle \{x\})$. Additionally, it is clear that $m(A_n \triangle \{x\}) \rightarrow 0$ as $n \rightarrow \infty$, so

$$d(A_n, \{x\}) \rightarrow 0 \text{ as } n \rightarrow \infty$$

It immediately follows from this result that $\{x\}$ is Lebesgue measurable. In fact, more is true: For each $n \in \mathbb{Z}_+$, $A_n \in \mathcal{E}$ and is an open cover of $\{x\}$, which shows

$$m^*(\{x\}) \leq m(A_n).$$

$m(A_n) \rightarrow 0$ as $n \rightarrow \infty$, so $m^*(\{x\}) = 0$, as was to be shown.

The result for singletons can be used to show that every countable subset of $\mathbb{R}$ is Lebesgue measurable with Lebesgue measure 0: if $U = \{x_1, x_2, \dots\} \subset \mathbb{R}$ is any countable subset, U , being a countable (disjoint) union of singletons, is Lebesgue measurable because $\mathcal{M}(m)$ is a σ -algebra. Furthermore, since m^* is countably additive on $\mathcal{M}(m)$,

$$m^*(U) = \sum_{n \in \mathbb{Z}_+} m^*(\{x_n\}) = 0$$

(b)

Show directly from the definition of the Lebesgue measure that $m^*(\mathbb{Q}) = 0$.

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 Fix $\epsilon > 0$ and write $\mathbb{Q} = \{q_1, q_2, \dots\}$. For each $n \in \mathbb{Z}_+$, let

$$I_n = (q_n - \frac{\epsilon}{2^{n+1}}, q_n + \frac{\epsilon}{2^{n+1}})$$

Each I_n has measure $m(I_n) = \frac{\epsilon}{2^n}$. $(I_n)_{n \in \mathbb{Z}_+}$ form an open cover for $\mathbb{Q}$, and

$$\sum_{n=1}^{\infty} m(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon,$$

as was to be shown.

(c)

Show that $m^*(\mathbb{R} \setminus \mathbb{Q}) = \infty$ and $m^*(\mathbb{Q} \cap [0, 1]) = 1$.

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 Since $\mathcal{M}(m)$ is a σ -algebra and $\mathbb{Q}$ is Lebesgue measurable, $\mathbb{R} \setminus \mathbb{Q}$ is also Lebesgue measurable. Recall also that σ -algebras always contain the ambient set, so $\mathbb{R}$ is Lebesgue measurable. Because m^* is monotonic on $\mathcal{M}(m)$ and there are bounded intervals of arbitrary length, it is clear that $m^*(\mathbb{R}) = \infty$. Using the countable additivity of m^* on $\mathcal{M}(m)$,

$$m^*(\mathbb{R} \setminus \mathbb{Q}) = m^*(\mathbb{R}) - m^*(\mathbb{Q}) = \infty,$$

as was to be shown.

$[0, 1]$ and $\mathbb{Q}$ are Lebesgue measurable, so that $\mathcal{M}(m)$ is a σ -algebra means $[0, 1] \setminus \mathbb{Q}$ and $[0, 1] \cap \mathbb{Q}$ are also Lebesgue measurable. The quick computations

$$\begin{aligned} m^*[0, 1] &= m[0, 1] = 1 \\ m^*(\mathbb{Q} \cap [0, 1]) &= 0 \text{ (by monotonicity)} \\ ([0, 1] \setminus \mathbb{Q}) \sqcup ([0, 1] \cap \mathbb{Q}) &= [0, 1] \end{aligned}$$

give

$$m^*([0, 1] \setminus \mathbb{Q}) = m^*[0, 1] - m^*(\mathbb{Q}) = 1.$$

2.**(a)**

Find a Lebesgue measurable set A such that $m(\overline{A}) > m(A)$.

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 Take $A = \mathbb{Q}$. $\overline{\mathbb{Q}} = \mathbb{R}$, and from Q1,

$$m(\mathbb{Q}) = 0 < m(\overline{\mathbb{Q}}) = \infty$$

(b)

Find a Lebesgue measurable set A such that $m(A^\circ) < m(A)$.

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 Let $A = \mathbb{R} \setminus \mathbb{Q}$, the set of irrational numbers. Since $\mathbb{Q}$ is measurable and measurable sets of a σ -algebra, A is also measurable. Further, since m is countably additive on measurable sets, by Q1, $m(A) = \infty$. $\mathbb{Q}$ is dense in $\mathbb{R}$ means $A^\circ = \emptyset$, and the fact $m(\emptyset) = 0$ means A satisfies the desired properties.

3.**(a)**

Prove that if $A, B \in \mathcal{E}$, then $\mu(B) = \mu(A \cap B) + \mu(B \setminus A)$.

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 As Brandon noted, if $A, B \in \mathcal{E}$, then $A \cap B \in \mathcal{E}$, which also means $B \setminus A \in \mathcal{E}$. μ is finitely additive on $\mathcal{E}$ and $B = (B \setminus A) \sqcup (A \cap B)$, giving

$$\mu(B) = \mu(A \cap B) + \mu(B \setminus A)$$

(b)

Prove that if $B \subset \mathbb{R}^p$ and $A \in \mathcal{E}$, then $\mu^*(B) = \mu^*(B \setminus A) + \mu^*(B \cap A)$.

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 Let's first show $\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B)$. Fix $\epsilon > 0$. There is a collection $(A_n)_{n \in \mathbb{Z}_+}$ such that

$$\text{each } A_n \in \mathcal{E} \text{ is open, } B \subset \bigcup_{n=1}^{\infty} A_n, \text{ and } \sum_{n=1}^{\infty} \mu(A_n) < \mu^*(B) + \epsilon$$

Since μ is regular, for each $n \in \mathbb{Z}_+$, pick $F_n, G_n \in \mathcal{E}$ such that G_n contains A and is open, F_n is contained in A and is closed, and

$$\mu(A) - \epsilon/2^n < \mu(F_n) \leq \mu(A) \leq \mu(G_n) < \mu(A) + \epsilon/2^n.$$

Recall that intersection of elementary sets is also elementary. By construction of the collections (E_n) , (G_n) , and (F_n) ,

$$\begin{aligned} \bigcup_{n=1}^{\infty} (A_n \cap G_n) &\text{ is an open cover for } B \cap A, \\ \bigcup_{n=1}^{\infty} (A_n \setminus F_n) &\text{ is an open cover for } B \setminus A \end{aligned}$$

Also, for each $n \in \mathbb{Z}_+$, the following holds by the construction of G_n and the monotocity of μ :

$$\begin{aligned} \mu(A_n \cap G_n) &= \mu(A_n) + \mu(G_n) - \mu(A_n \cup G_n) \\ &< \mu(A_n) + \mu(A) + \epsilon/2^n - \mu(A_n \cup A) \\ &= \mu(A_n \cap A) + \epsilon/2^n \end{aligned}$$

Hence

$$\begin{aligned}\mu^*(B \cap A) &\leq \sum_{n=1}^{\infty} \mu(A_n \cap G_n) \\ &< \sum_{n=1}^{\infty} \mu(A_n \cap A) + \epsilon\end{aligned}$$

Similarly,

$$\begin{aligned}\mu(A_n \setminus F_n) &= \mu(A_n \cup F_n) - \mu(F_n) \\ &< \mu(A_n \cup A) - \mu(A) + \epsilon/2^n \\ &= \mu(A_n \setminus A) + \epsilon/2^n,\end{aligned}$$

which gives

$$\mu^*(B \setminus A) < \sum_{n=1}^{\infty} \mu(A_n \setminus A) + \epsilon$$

Finally, since $(A_n \setminus A) \sqcup (A_n \cap A) = A_n$ for each n ,

$$\begin{aligned}\mu^*(B \setminus A) + \mu^*(B \cap A) &< \sum_{n=1}^{\infty} \mu(A_n) + 2\epsilon \\ &< \mu^*(B) + 3\epsilon\end{aligned}$$

Letting $\epsilon \rightarrow 0$, $\mu^*(B \setminus A) + \mu^*(B \cap A) \leq \mu^*(B)$.

Next, let's show $\mu^*(B) \leq \mu^*(B \cap A) + \mu^*(B \setminus A)$. Fix $\epsilon > 0$. There are collections $(A_n)_{n \in \mathbb{Z}_+}, (B_n)_{n \in \mathbb{Z}_+} \subset \mathcal{E}$ such that each A_n and B_n are open and

$$\begin{aligned}B \setminus A &\subset \bigcup_{n=1}^{\infty} A_n, & \sum_{n=1}^{\infty} \mu(A_n) &< \mu^*(B \setminus A) + \epsilon \\ B \cap A &\subset \bigcup_{n=1}^{\infty} B_n, & \sum_{n=1}^{\infty} \mu(B_n) &< \mu^*(B \cap A) + \epsilon\end{aligned}$$

For each $n \in \mathbb{Z}_+$, since μ is regular, there are also $F_n, G_n \in \mathcal{E}$ such that G_n contains A and is open, F_n is contained in A and is closed, and

$$\mu(A) - \epsilon/2^{n+1} < \mu(F_n) \leq \mu(A) \leq \mu(G_n) < \mu(A) + \epsilon/2^{n+1}.$$

For each $n \in \mathbb{Z}_+$, $A_n \setminus F_n$ and $B_n \cap G_n$ are still both elementary. Further, it is clear that $\bigcup_{n=1}^{\infty} (A_n \setminus F_n)$ is an open cover for $B \setminus A$ and $\bigcup_{n=1}^{\infty} (B_n \cap G_n)$ is an open cover for $B \cap A$, so

$$\bigcup_{n=1}^{\infty} ((A_n \setminus F_n) \cup (B_n \cap G_n)) \text{ is an open cover for } B$$

For each $n \in \mathbb{Z}_+$, $(A_n \setminus F_n) \cap (B_n \cap G_n) \subset G_n \setminus F_n$, and $\mu(G_n \setminus F_n) = \mu(G_n) - \mu(F_n) < \epsilon/2^n$. Using the monotocity of μ ,

$$\begin{aligned}\mu((A_n \setminus F_n) \cup (B_n \cap G_n)) &\geq \mu(A_n \setminus F_n) + \mu(B_n \cap G_n) - \mu(G_n \setminus F_n) \\ &\geq \mu(A_n \setminus F_n) + \mu(B_n \cap G_n) - \epsilon/2^n\end{aligned}$$

Putting all the information together and once again using the monotonicity of μ ,

$$\mu^*(B) \leq \sum_{n=1}^{\infty} ((A_n \setminus F_n) \cup (B_n \cap G_n))$$

$$\sum_{n=1}^{\infty}$$

(c)

4.

Let X be a set and $\mathcal{M} \subset 2^X$ be a σ -algebra.

(a)

Take another set Y and a function $h : X \rightarrow Y$. Show $\mathcal{M}' = \{B \subset Y \mid h^{-1}(B) \in \mathcal{M}\}$ is a σ -algebra on Y .

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Take $B \in \mathcal{M}'$. Since $\mathcal{M}$ is a σ -algebra, $X \setminus h^{-1}(B) \in \mathcal{M}$.

$$h^{-1}(Y \setminus B) = X \setminus h^{-1}(B),$$

so $Y \setminus B \in \mathcal{M}'$.

Take $(B_n)_{n \in \mathbb{Z}_+} \subset \mathcal{M}'$. Then

$$\bigcup_{n=1}^{\infty} h^{-1}(B_n) = h^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right) \in \mathcal{M},$$

so $\bigcup_{n=1}^{\infty} B_n \in \mathcal{M}'$. This shows $\mathcal{M}'$ is a σ -algebra on Y .